

CrossHAR: Generalizing Cross-dataset Human Activity Recognition via Hierarchical Self-Supervised Pretraining

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The increasing availability of low-cost wearable devices and smartphones has significantly advanced the field of sensor-based human activity recognition (HAR), attracting considerable research interest. One of the major challenges in HAR is the domain shift problem in cross-dataset activity recognition, which occurs due to variations in users, device types, and sensor placements between the source dataset and the target dataset. Although domain adaptation methods have shown promise, they typically require access to the target dataset during the training process, which might not be practical in some scenarios. To address these issues, we introduce CrossHAR, a new HAR model designed to improve model performance on unseen target datasets. CrossHAR involves three main steps: (i) CrossHAR explores the sensor data generation principle to diversify the data distribution and augment the raw sensor data. (ii) CrossHAR then employs a hierarchical self-supervised pretraining approach with the augmented data to develop a generalizable representation. (iii) Finally, CrossHAR fine-tunes the pretrained model with a small set of labeled data in the source dataset, enhancing its performance in cross-dataset HAR. Our extensive experiments across multiple real-world HAR datasets demonstrate that CrossHAR outperforms current state-of-the-art methods by 10.83% in accuracy, demonstrating its effectiveness in generalizing to unseen target datasets.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**.

Additional Key Words and Phrases: Human activity recognition, Cross-dataset, Cross-domain, Self-supervised learning

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1 INTRODUCTION

Recent advancements in human activity recognition (HAR) have significantly enhanced practical applications in diverse fields such as healthcare [26] and manufacturing [54, 81]. The widespread use of wearable devices

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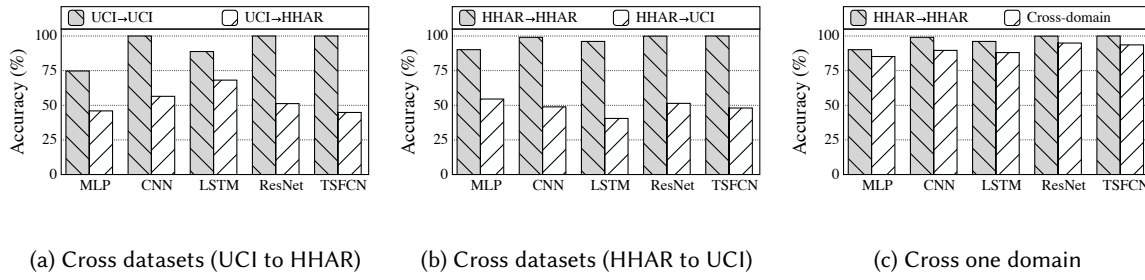


Fig. 1. Comparison of performance drop between (a)(b) **cross-dataset** and (c) **cross-domain** settings. Results highlight that the cross-dataset setting brings more performance drop.

and smartphones, equipped with Inertial Measurement Units (IMU), has led to increased attention in HAR from both research and industry sectors [8, 61]. These devices facilitate the recognition of activities like walking and downstairs at a low cost and with broad coverage. A primary challenge in HAR is the issue of domain shift, where discrepancies in the data distribution between training and testing datasets adversely affect recognition performance. The term "domain" refers to factors influencing sensor readings, such as user groups and sensor placement. Domain shift predominantly occurs in two scenarios: **cross-domain** [25, 50, 52, 53] and **cross-dataset** [51]. **Cross-domain activity recognition** involves scenarios where training and testing data originate from different domains within the same dataset. For example, training data might come from user A and testing data from user B, both sourced from the same dataset [77]. However, real-world scenarios often present more complex challenges than those found in cross-domain settings. This leads to **cross-dataset activity recognition**, where models are trained on one dataset (the *source dataset*) and evaluated on another (the *target dataset*). The *source* and *target datasets* are typically collected by different research groups, each following their own data collection protocols. This results in variations across multiple domains, such as user groups, device types, sensor placements, and data collection guidelines [56, 65]. Consequently, the *cross-dataset* setting involves heterogeneous domain shifts, posing a more complex challenge in the context of HAR.

Due to the challenges of the cross-dataset activity recognition, we suppose there will be a significant performance drop when transferring from the *source dataset* to the *target dataset*. To verify our hypothesis, we conducted an empirical investigation on two widely investigated datasets, UCI [56] and HHAR [65] datasets, with five popular activity recognition models. The average accuracy of cross-dataset (trained with one source dataset) drops 42.39% while cross-one-domain (trained with 40% individuals) drops 6.77%, as shown in Fig. 1a and Fig. 1b. Compared to the cross-domain setting in Fig. 1c, the cross-dataset setting experiences a significantly greater decrease in recognition accuracy due to multiple domain shifts existing simultaneously, which further demonstrates the challenge of cross-dataset activity recognition.

To mitigate the challenges brought by domain shifts in HAR, a variety of methods have been proposed. While only a few specifically target cross-dataset HAR, certain relevant concepts from these methods can be adapted [42, 51]. *Domain adaptation*, a widely studied technique to address the domain shift issue [25, 34, 80, 80], aims to bridge the gap between source and target domains by incorporating data from both during the training phase. This approach can enhance model performance by aligning the representation spaces of the different domains. However, this method often assumes the availability of target domain data during training, which may not be practical in many real-world scenarios [42]. In practice, models are frequently required to generalize across multiple new environments without prior exposure to data from these settings [53]. For instance, a model trained on data from a wrist-worn device on a student may need to be deployed for an elderly individual carrying a smartphone in their front pocket. Given these practical constraints, *domain generalization* [8, 35, 76, 84, 89],

focuses on training models exclusively with source domain data to achieve robust performance on unseen target domains, becomes a more promising approach. This approach is accomplished by learning domain-invariant representations that are effective across various domains. However, most existing domain generalization methods are either not specifically designed for IMU sensor data or do not fully leverage the unique characteristics of IMU data. Additionally, the potential of using unlabeled data to enhance the generalization capabilities of these models also remains to be explored.

Therefore, we explore two opportunities to further improve the performance: *accumulated unlabeled sensor data* and *self-supervised pretraining models*. (i) The accumulated large-scale unlabeled sensor data has recently provided a unique opportunity for addressing the generalization issue of activity recognition models. The rich information embedded within unlabeled sensor data can probably enhance the generalization capabilities of human activity recognition models [25]. By incorporating this unlabeled data into our training process, we can potentially unveil latent patterns and representations that are transferable across unseen domains. (ii) Furthermore, recent advancements in pretraining methodologies in the computer vision [11, 19] and natural language processing (NLP) [32] communities have shown remarkable effectiveness in improving generalization [31, 76]. These models are pretrained with self-supervised losses on large-scale unlabeled datasets. Even with good performance, most pretraining methods are not designed for IMU sensor data or not designed for the generalization setting [83]. Few studies have shed light on the important problem of whether and how unlabeled data benefit the generalization capability of sensor-based HAR models [25, 48].

In this paper, to improve the performance of cross-dataset HAR only with the data from the source dataset, we present CrossHAR, a novel **C**ross-dataset **H**uman **A**ctivity **R**ecognition model that leverages unlabeled sensor data and the sensor data generation principles. Because CrossHAR has no prior knowledge about the distribution of the target dataset, we carefully design three components to increase the generalization capability. (i) *Physically-informed Sensor Data Augmentation*. Based on a systematic analysis of the origin of the performance gap between the source dataset and the target dataset, we design a physically-informed augmentation algorithm to improve the generalization capability of models trained on the source dataset. (ii) *Hierarchical Sensor Data Pretraining*. To utilize the unlabeled data, we design a local-global hierarchical self-supervised pretraining model. Specifically, we design a masked sensor modeling approach to capture the local pattern of sensor data. Moreover, we design a contrastive pertaining objective to capture the global pattern by regularizing the pretraining process. This hierarchical pretraining paradigm enables the model to capture domain-invariant sensor patterns from unlabeled data. (iii) *Fine-Tune Activity Recognition*. Finally, we utilize a small amount of labeled data from the source dataset to fine-tune the pretrained model for activity recognition on the target dataset. In summary, this paper makes the following three contributions.

- To our knowledge, this is the first study that aims to address the cross-dataset HAR problem from the perspective of self-supervised pretraining. We design CrossHAR, a novel HAR model that learns generalizable representations via physically-informed sensor augmentation and hierarchical sensor data pretraining design. CrossHAR only requires unlabeled data and a small set of labeled data purely from the source dataset during the model training phase.
- We evaluate CrossHAR on four real-world datasets compared to multiple SoA models. Results show that CrossHAR outperforms SoA models by 10.83% in accuracy in cross-dataset HAR, which sheds light on the utility of physically-informed augmentation and sensor data pretraining mechanisms for the generalization of activity recognition.
- Moreover, we deploy the pretrained CrossHAR on a smartphone to further conduct a case study for cross-dataset activity recognition. The average inference speed is 0.0225 seconds on a Redmi smartphone, which demonstrates the high efficiency of CrossHAR for real-world usage.

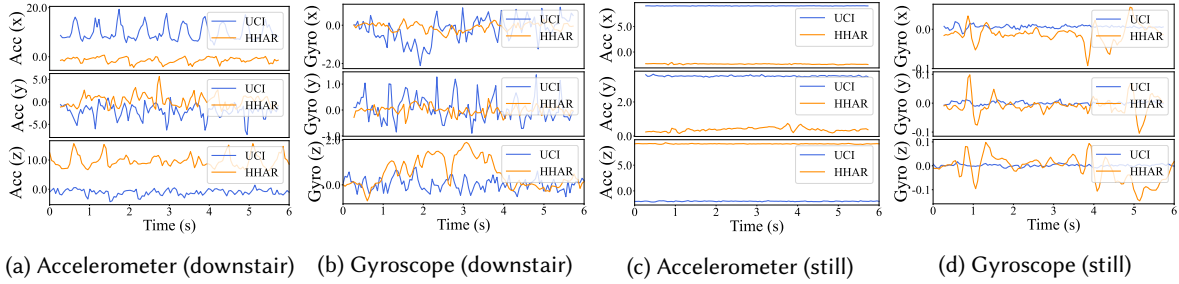


Fig. 2. Illustration of cross-dataset domain shifts. For the same activity, values of different channels in the accelerometer sensor and gyroscope sensor vary significantly.

2 PRELIMINARIES AND PROBLEM DEFINITION

In this section, we introduce the background of cross-dataset HAR compared to the cross-domain setting. Then, we introduce different model training paradigms and give a formal problem definition.

2.1 Cross dataset and cross domain

Domain shift is one of the main challenges in HAR. We summarize the domain shift problems into **cross domain** and **cross dataset** according to existing studies [8, 51, 77]. We discuss the differences between them and present why cross dataset is more challenging.

Cross domain. *Cross-domain HAR* means the training data and testing data are from different domains of the same dataset. Thus, the challenge is mainly brought by domain shift in one domain, and across-domain evaluations are conducted to test the model performance across one domain, e.g., across user groups [25, 50, 52]. However, in the real-world scenario, the assumption that only one domain is different between source and target datasets does not always hold, e.g., both user group and device placement are different.

Cross dataset. *Cross-dataset HAR* means models are trained on one dataset (*source dataset*) and evaluated on another dataset (*target dataset*) [84]. Different from cross-domain settings, cross-dataset is more challenging, because the data in the source dataset are different from the target dataset in multiple domains simultaneously, such as user groups, device placement, and device types. For example, in UCI [56] and HHAR [65] datasets, users, device placements, device types, sampling rates, and more importantly, the data collection processes differ from each other significantly. Note that if only one domain is different between two datasets, the *cross-dataset* is the same as *cross-domain*. Fig. 2 illustrates such multi-domain shift in both accelerometer and gyroscope sensors. Therefore, the distribution of IMU data is significantly different for the same category of activities.

We represent the source dataset as D_{source} and the target dataset as D_{target} . Unlike most existing *cross-domain* studies that focus on domain shift caused by one domain [25, 50, 52], we aim to generalize from D_{source} to D_{target} , addressing the shift brought by the combination of multiple domains. To address such a strong domain shift, we can design different model training paradigms, which are introduced in the next subsection.

2.2 Model training paradigm

We summarize model training paradigms into three main categories according to the source of training data. Note that they include both supervised training and self-supervised training models.

Paradigm I: Training with data from the source dataset (D_{source}) and fine-tuning with a subset data from the target dataset (D_{target}) [8, 57, 59]. This paradigm is widely studied and can achieve good performance on the target dataset due to the model ($\mathcal{M}$) can capture the feature and label distribution of the target dataset.

Table 1. Mathematical notations.

Symbol	Description
x_i	Input IMU sensor data matrix, consisting of accelerometer reading and gyroscope reading.
D_{source}	Source dataset for pretraining and fine-tuning.
D_{target}	Target dataset for model evaluation.
L	Length of IMU data in a sample.
N	Number of samples in a batch during pretraining.
$\mathcal{M}(., \Theta)$	Pretrained model with trainable parameters Θ .
$\mathcal{M}(., \Phi)$	Fine-tuned model with trainable parameters Φ .
A	Transformation matrix for physically-informed augmentation.
η	Learning rate for model training.
$\mathcal{L}_m$	Loss function of masked sensor modeling for self-supervised model pretraining.
$\mathcal{L}_r$	Loss function of contrastive regularization for self-supervised model pretraining.
$\mathcal{L}$	Overall loss function for model pretraining.
$\mathcal{L}_{classification}$	Loss function of activity recognition for model fine-tuning.
α	Weight of the masked sensor modeling loss $\mathcal{L}_m$.
β	Weight of the contrastive regularization loss $\mathcal{L}_r$.

$$\text{Paradigm I : } D_{source} + D_{target} \rightarrow \mathcal{M} \quad (1)$$

Paradigm II: Training with data from source dataset (D_{source}) and unlabeled data from target dataset (D_{target}^U) [25, 47, 48]. This training paradigm is more challenging due to no labeled data from the target dataset is available during the model training process. However, it still achieves good performance due to the unlabeled data providing distribution information of the target dataset. Models trained with unlabeled target data and labeled source data can align the distribution from different datasets in the representation space.

$$\text{Paradigm II : } D_{source} + D_{target}^U \rightarrow \mathcal{M} \quad (2)$$

Paradigm III: Training only with data from the source dataset (D_{source}) [50, 52]. This paradigm is less studied and more challenging compared to the first two. However, this is more practical for real-world applications since the model training environment is normally different from the real deployment environment. Moreover, a trained model is supposed to achieve good performance in different environments with diverse distribution shifts.

$$\text{Paradigm III : } D_{source} \rightarrow \mathcal{M} \quad (3)$$

Due to Paradigm III having practical usage in real-world deployment and being less studied, we aim to design a framework within Paradigm III, for generalizable HAR. The detailed problem definition is given below.

2.3 Problem definition

Given a source dataset D_{source} containing a collection of IMU samples (x_i) with corresponding human activity labels (y_i), $D_{source} = \{(x_i, y_i)\}_{i=1}^{n_s}$, we aim to learn a model $\mathcal{M}(., \Phi): \mathcal{X} \rightarrow \mathcal{Y}$, which can generalize to unseen target dataset $D_{target} = \{(x_i, y_i)\}_{i=1}^{n_t}$ for human activity recognition. n_s and n_t are the number of samples in D_{source} and D_{target} . $\mathcal{X}$ is the input space, $\mathcal{Y}$ is the label space including various activities such as *walking*. We assume they are identical in D_{source} and D_{target} . Note that, the target dataset D_{target} is only available during the testing phase.

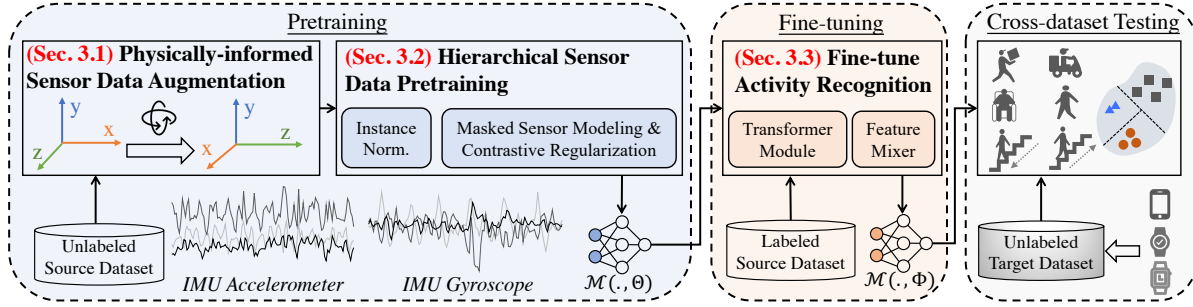


Fig. 3. Illustration of CrossSHAR (Sec. 3) and the cross-dataset testing process (Sec. 4). CrossSHAR consists of three concrete components as in Sec. 3.1, Sec. 3.2, and Sec. 3.3. The pretraining phase generates the pretrained model $\mathcal{M}(\cdot, \Theta)$. The fine-tuning phase generates the fine-tuned model $\mathcal{M}(\cdot, \Phi)$ for cross-dataset activity recognition.

Specifically, D_{source} consists of a labeled subset D_{source}^L and an unlabeled subset D_{source}^U . D_{source}^U is utilized for the self-supervised pretraining and D_{source}^L is involved in the supervised fine-tuning process. The self-supervised pretraining generates the pretrained model $\mathcal{M}(\cdot, \Theta)$, and finally, we obtain the fine-tuned model $\mathcal{M}(\cdot, \Phi)$ after the fine-tuning process. $D_{source}^L = \{(x_i, y_i)\}_{i=1}^{n_l}$, $D_{source}^U = \{(x_i, y_i)\}_{i=n_l+1}^{n_l+n_u}$, $n_s = n_l + n_u$, n_l is the length of the labeled subset D_{source}^L and n_u is the length of the unlabeled subset D_{source}^U . We focus on the low resource setting, where $n_l < n_u$, e.g., $n_l : n_u = 1 : 9$. To enhance the readability of this paper, we summarize important notations in Table 1.

3 FRAMEWORK

Our model design is driven by two key intuitions from IMU data and self-supervised pretraining. Firstly, one of the main reasons that decreases HAR performance is the domain shift in IMU data. Device placement and orientations play a vital role that impacting the IMU data distribution. Therefore, we can mitigate this shift if we can understand the data generation process and augment IMU data in diverse data collection environments. Secondly, drawing inspiration from advancements in computer vision and natural language processing, self-supervised learning methods have shown considerable promise in improving model generalization capabilities [31, 76]. Therefore, we can probably address the domain shift issue by properly designing a pretraining method for IMU data. According to these intuitions, we design a novel model CrossSHAR to improve cross-dataset HAR. As shown in Fig. 3, CrossSHAR consists of the following three modules.

(i) Physically-informed Sensor Data Augmentation. The first component of CrossSHAR is physically-informed sensor data augmentation. By analyzing the IMU data from multiple datasets, we identify the significant distribution discrepancy among sensor readings. Then, we design an augmentation approach to augment the data in the source dataset, motivated by the data generation principles of accelerometer and gyroscope sensors.

(ii) Hierarchical Sensor Data Pretraining. The second component of CrossSHAR involves pretraining based on masked sensor modeling and contrastive regularization. Firstly, we utilize the structure of language models to identify and learn local features from the IMU sensor data, thereby improving the understanding of the inherent patterns and relationships within IMU data. Subsequently, we incorporate a contrastive regularization module to further improve the self-supervised learning process and the model's generalization capability.

(iii) Fine-tune Activity Recognition. The third component of CrossSHAR involves the supervised training with data only from the source dataset. Specifically, after the pretraining phase, CrossSHAR is further fine-tuned through a lightweight supervised classification layer, with a limited set of labeled data, enabling the model to recognize

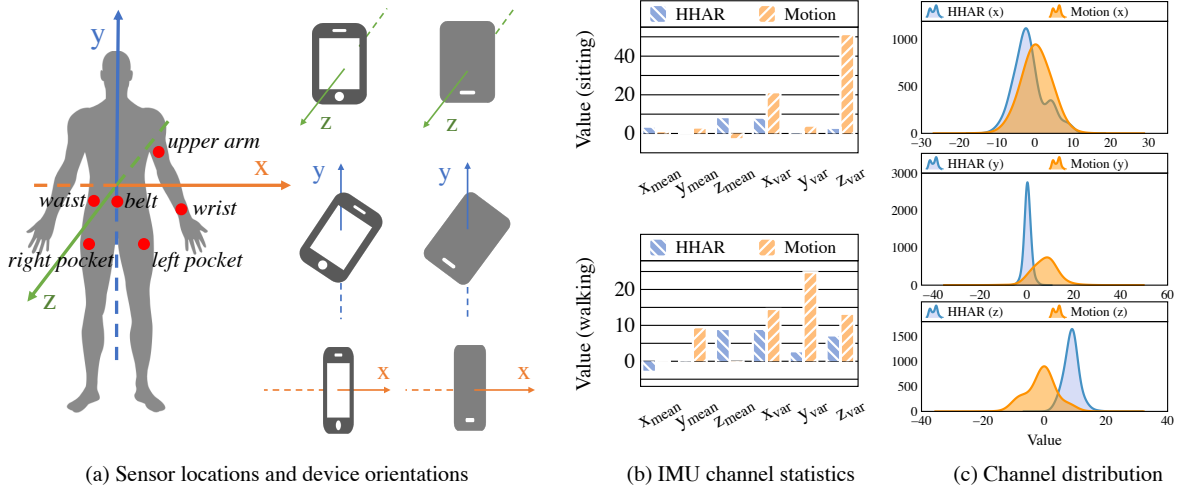


Fig. 4. Illustration of smartphone sensor orientations. (a) presents the coordinate system according to the user and sensor placement on the user's body. We also show the smartphone orientations relative to the user coordinate system. (b) presents the IMU distribution heterogeneity with an example of HHAR and Motion datasets. (c) presents the channel distribution variance of 3-channel accelerometer readings of two datasets.

and classify different activities accurately. The fine-tuning phase is important in aligning the pretrained model with the specific task of HAR.

3.1 Physically-informed sensor data augmentation

3.1.1 Motivation. One of the main reasons that causes the performance drop when transferring from the source to the target dataset is the heterogeneous device orientations, resulting from potential factors like device placements and gestures of users when holding mobile devices. Specifically, IMU data consists of sensor readings from three axes, x , y , and z , to represent the direction and magnitude of acceleration and gyroscope. These values are directly influenced by the orientation of the smartphone screen [1], which may randomly generate different orientations relative to the user due to user habits and device positioning. Therefore, for the same activity, the distribution of magnitude on the three axes can vary significantly. As an example, we present different smartphone orientations relative to the user in Fig. 4(a). Even though for most public datasets, the data within one dataset have a similar distribution in three channels. For different datasets, the channel distributions vary significantly due to the uncertainty during data collection processes. We illustrate this phenomenon in Fig. 4(b), where for both *sitting* and *walking*, the highest value in the Motion dataset [46] is on the y -axis, while it is on the z -axis in the HHAR dataset [65]. For variance, Motion is significantly higher than HHAR. Therefore, the models trained on the source dataset will not perform well on the target datasets due to the significant distribution shifts.

3.1.2 Algorithm design. To mitigate the distribution shift, we design a sensor augmentation algorithm to increase the model's generalization ability. Intuitively, in three-dimensional space, the orientation of a mobile phone, i.e., the direction in which the phone screen faces, can be at any angle as in Fig. 4(a), thereby values in three channels change with the device orientations. Based on this observation, we argue that by properly simulating the device moving process, we can increase the data diversity and improve the model's generalization ability. Motivated by rotation-based augmentation on the accelerometer sensor [6, 74], we design coordinate transformations to obtain sensor readings from various orientations. Specifically, we rotate channels of both the accelerometer

and gyroscope simultaneously, i.e., the rotation angle for the same channel in the accelerometer sensor and the gyroscope sensor is the same. Note that, the number of generated IMU samples can be infinite due to the infinite space of device orientations [74]. However, to increase the model training efficiency and reduce data redundancy, we augment IMU data by generating six mutually orthogonal directions, which can better represent all possible orientations of smartphones with fewer samples, instead of randomly selecting a large number of directions. Given the current device orientation (x,y,z) , there are six possible permutations of the axis order, i.e., (x,y,z) , (x,z,y) , (y,x,z) , (y,z,x) , (z,x,y) , and (z,y,x) , which represent six rotation angles. The same permutation of axes is applied to both the accelerometer and gyroscope data. This means that each original sensor data sample is expanded to six samples that are orthogonal to each other.

In our setting, an IMU sample x_i consists of an accelerometer sensor Acc_i and a gyroscope sensor $Gyro_i$, both consisting of three channels. $x_i = [Acc_i^x \ Acc_i^y \ Acc_i^z \ Gyro_i^x \ Gyro_i^y \ Gyro_i^z]$. We take the accelerometer ($Acc_i = [Acc_i^x \ Acc_i^y \ Acc_i^z]$) as an example to present our approach because the gyroscope follows the same channel distribution as the accelerometer. Then, we design the transformation matrix A , a 3×3 matrix to transfer the original sensor data by swapping the order of three channels:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (4)$$

a_{rc} is the value in A , $r \in \{1, 2, 3\}$, $c \in \{1, 2, 3\}$, $a_{rc} \in \{0, 1\}$. By multiplying Acc_i with the transformation matrix A , we obtain the transformed or augmented sensor data. An example of Acc_i^{aug} is as follows:

$$Acc_i^{aug} = [Acc_i^x \ Acc_i^z \ Acc_i^y] \leftarrow Acc_i \cdot A = [Acc_i^x \ Acc_i^y \ Acc_i^z] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (5)$$

The generation of augmented gyroscope data $Gyro_i^{aug}$ follows the same transformation with Acc_i . By combining the transformed accelerometer Acc_i^{aug} and gyroscope $Gyro_i^{aug}$, we obtain the transformed IMU data x_i^{aug} . In total, the number of IMU samples is increased to six times the original size. For simplicity, we still use x_i to represent IMU samples in the augmented dataset. According to our simulation of the device orientations, the augmented data captures diverse data collection scenarios and thereby potentially benefits the HAR model.

3.2 Hierarchical sensor data pretraining

Given the physically-informed augmented sensor data, the next step is to design a self-supervised learning algorithm to learn sensor representations. Intuitively, both local patterns and global patterns of IMU sensor data benefit HAR. *Local patterns* refer to specific short-term patterns or behaviors within IMU samples. Local patterns include instantaneous changes in orientation, abrupt accelerations, rotational rates, etc., capturing specific actions or events that occur within a short time frame, such as device rotations. On the other hand, *global patterns* reflect the overall trends of the IMU sensor data. Therefore, we design a masked sensor modeling pretraining approach to capture the local sensor patterns, followed by a contrastive regularization pretraining approach to model the global sensor patterns. Through this hierarchical pretraining design, we can better capture the motion and behavior of the IMU sensor in the representation space and improve cross-dataset HAR.

3.2.1 Masked sensor modeling pretraining. Even though our physically-informed augmentation mitigates the distribution shift among different channels of samples from source and target datasets. There are still distribution discrepancies in each channel as shown in Fig. 4(c), where the channel y of the accelerometer in Motion dataset is significantly higher than in HHAR. Thus, directly pretrain on the source dataset still faces a significant distribution

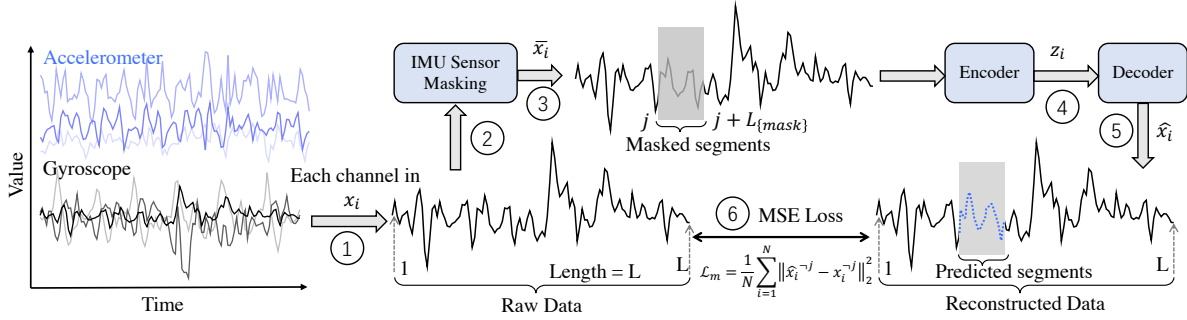


Fig. 5. Illustration of sensor masking and masked sensor prediction. Note that we use one channel to present the masking process, all channels in each IMU sample x_i share the same process and are masked at the same location j . ⑥ calculates the reconstruction loss $\mathcal{L}_m$ between Raw Data with the Reconstructed Data.

shift when applied to the target dataset. Therefore, we might further improve the performance by normalizing the sensor data to align their distribution in each channel.

Sensor data normalization. Intuitively, the statistical properties of sensor data change over time and data collection environment, e.g., device sensor properties. Therefore, the sensor data distribution in the target dataset might be different from the source dataset significantly, which will probably decrease the generalization performance [70]. In time series forecasting, reversible instance normalization [36] is designed to mitigate domain shift. The reversible instance normalization removes and restores the statistic information for a time series instance to reduce the distribution shift across multiple domains. Different from the forecasting task, we adapt the normalization only in the input phase to align data distributions across source domains and target domains. We assume that from the shape or trend of the sensor data, we can distinguish different activities. Therefore, the normalization would not bring a negative impact. Specifically, following existing studies [70, 73], the instance normalization is applied to each channel (axis) dimension separately, avoiding destroying original channel-specific properties and stabilizing the learning process [36]. This involves normalizing each data point by subtracting its mean and dividing it by the standard deviation.

Sensor masking. Masked language modeling has been widely adopted by models based on BERT [32] and Transformer [75]. Masked language modeling masks one token for each subsequence of textual data. However, given the continuous characteristics of IMU data, multiple points in each dimension are sampled each second. Masking one point each time makes the pretraining task too easy and the pretraining model cannot learn meaningful patterns of IMU data. Thus, motivated by LIMUBERT [83], we mask multiple IMU segments during pretraining. Given an IMU sample x_i , a segment of sensor data is denoted by x_i^j , the segment within x_i starting from j with a defined length L_{mask} , $1 \leq j \leq L - L_{mask}$, $1 \leq L_{mask} \leq L$, $x_i \in \mathbb{R}^{L \times D}$. L is the length of an IMU sample and D is the dimension or the number of channels of an IMU sample. For example, D is 6 in the UCI dataset [56]. As shown in steps ① ② ③ of Fig. 5, for each channel of x_i , we mask multiple segments rather than several points each time. The masked segments x_i^j are replaced as zero. Similar masking strategies have also been explored in other time series data [12, 13]. The sensor masking operation $\text{SensorMask}(\cdot)$ is shown below.

$$\bar{x}_i = \text{SensorMask}(x_i) \quad (6)$$

As shown in step ③ and ④ of Fig. 5, after the sensor masking process, we have a new batch of samples $\bar{x}_i^j$. An $\text{Encoder}(\cdot)$ is designed based on Multi-head Attention in the BERT model [32]. The masked samples are thus transferred to latent representations z_i as:

$$z_i = \text{Encoder}(\bar{x}_i) \quad (7)$$

Based on the latent representation z_i , we design two sub-modules to guide the model to learn generalizable features via self-supervised pretraining. Firstly, a $\text{Decoder}(\cdot)$ is designed to decode z_i and predict the masked segments in each sample, as shown in step ⑤ Fig. 5.

$$\hat{x}_i = \text{Decoder}(z_i) \quad (8)$$

$\hat{x}_i$ is the reconstructed sample according to the prediction result. We implement the $\text{Decoder}(\cdot)$ based on BERT [32]. According to the original sample x_i and the reconstructed sample, we compute the difference by MSE loss [28], as shown in step ⑥ of Fig. 5. The loss $\mathcal{L}_m$ is utilized to update the model parameters in the pretraining step.

$$\mathcal{L}_m = \frac{1}{N} \sum_{i=1}^N \|\hat{x}_i^{-j} - x_i^{-j}\|_2^2 \quad (9)$$

where j represents the location of masked segments from the original sample, x_i^{-j} means calculating the reconstruction loss only on masked segments and ignoring others.

3.2.2 Contrastive regularization. The batch of latent representation $\{z_i\}_{i=1}^N$ also goes to another branch, i.e., our $\text{Regularization}(\cdot)$ module, which is implemented based on contrastive learning [11]. The motivation of this module lies in two aspects. Firstly, while the masked sensor modeling pretraining captures the local patterns of IMU data [83], the global patterns are also important because they reflect the overall trends and are less impacted by sudden sensor noises. Secondly, unsupervised contrastive learning has been proven to be effective in capturing domain-invariant global patterns for classification tasks in computer vision [11, 35] and natural language processing [67]. Therefore, we design a contrastive regularization module to regularize the self-supervised pretraining process for IMU data.

$$r_i = \text{Regularization}(z_i) \quad (10)$$

The core idea of unsupervised contrastive learning is learning by comparison utilizing the inherent characteristics of the raw data [11]. Firstly, given an original sample, positive and negative samples are generated according to data characteristics. By contrasting positive pairs (similar samples) against negative pairs (dissimilar samples), models learn to recognize and encapsulate the underlying similarities as well as differences in the data. The fundamental objective is to minimize the distance between positive pairs and maximize the distance between negative pairs in the representation space [11]. Thus, to effectively regularize model parameters in the self-supervised pretraining process, an important step is constructing appropriate positive and negative pairs utilizing the characteristics of IMU data. We present the construction process in Fig. 6.

As in Fig. 6(a), given an IMU sample x_i , we construct two views of x_i according to the characteristics of IMU data. In contrastive learning, data transformations are widely studied to encourage the model to learn robust representations [11, 15]. For our scenario, the measurement of IMU data is impacted by various factors like device placement and sensor hardware. Therefore, to increase the robustness of learned representation, we adapt the transformation approaches [11, 16, 79] and design h and h' . Specifically, h refers to scaling the original IMU sample and changing its measurement results to get the first augmented sample. Specifically, the scaling is achieved by the following process. Firstly, we generate a Gaussian distribution $\mathcal{N}(2, 1.1)$, the mean and standard deviation are chosen following existing studies [15]. Then, for each time step, we randomly sample a value from

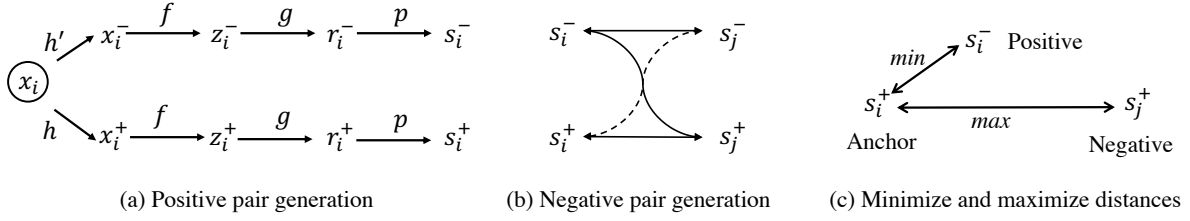


Fig. 6. Illustration of contrastive regularization.

this Gaussian distribution and multiply it by the values of all channels in the sensor data at that time step. h' refers to partitioning IMU data into multiple segments and shuffling the location of different segments with random noises added to get the second augmented sample. Thus, we have $2N$ samples (x_i^- and x_i^+) from the original batch with N samples.

In Fig. 6(a), f refers to the Encoder($\cdot$) in the masked sensor modeling process, which transforms x_i^- and x_i^+ into representation z_i^- and z_i^+ in the masked sensor modeling module. g is the Regularization($\cdot$) module, which transforms z_i^- and z_i^+ into latent representation r_i^- and r_i^+ with a Transformer model. The Regularization($\cdot$) module learns contextualized IMU representations, and comprehensively captures the dependencies and intricacies within segments of IMU data, thereby increasing the generalization of learned representation. Finally, a Projection($\cdot$) module p transforms r_i^- and r_i^+ into s_i^- and s_i^+ as the final representation. We implement p with a simple MLP module. Thus, we obtain the positive pairs $\langle s_i^-, s_i^+ \rangle$.

$$s_i = \text{Projection}(r_i) \quad (11)$$

The construction of negative pairs is shown in Fig. 6(b). We combine different samples to construct negative pairs, e.g., s_i^+ with s_j^- and s_j^+ , where $i \neq j$. For each sample s_i , we can construct $2(N - 1)$ negative pairs. Based on the positive pairs and negative pairs, we show the idea of contrastive learning for IMU data in Fig. 6(c). The augmented two views s_i^+ and s_i^- both serve as anchors. During the pretraining process, the model aims to maximize the distance between the anchor with negative samples (negative pairs) and minimize the distance with positive samples (positive pairs) [11]. Thus, the contrastive regularization loss is defined as follows:

$$\mathcal{L}_r = - \sum_{i=1}^N \log \frac{\exp(\text{sim}(s_i, s_i^+)/\tau)}{\sum_{j=1}^{2N} \mathbb{1}_{[j \neq i]} \exp(\text{sim}(s_i, s_j)/\tau)} \quad (12)$$

where sim is a similarity function measuring the distance between two representations, τ is a temperature parameter, s_i^+ denotes the augmented positive sample, and s_j denotes the augmented negative sample. $\mathbb{1}_{[j \neq i]} \in \{0, 1\}$ denotes an indicator function to filter the calculation when $j = i$.

3.2.3 Self-supervised pretraining. Given the masked sensor reconstruction loss and the contrastive regularization loss, we can update the model with unlabeled data from the source dataset D_{source} . The overall loss function is the combination of $\mathcal{L}_m$ and $\mathcal{L}_r$. Intuitively, we can follow the classic model optimization process and update the model with the overall loss $\mathcal{L}$.

$$\mathcal{L} = \alpha \cdot \mathcal{L}_m + \beta \cdot \mathcal{L}_r \quad (13)$$

where α and β represent the weights of two losses in the self-supervised pretraining process, respectively. However, we argue that the convergence speed of the two losses is different. Therefore, directly optimizing the model with $\mathcal{L}$ might not converge to a good position in the optimization space. During the experiments, we also observed similar phenomena (see Sec. 4.5.2). Therefore, we design a sequential updating paradigm for the

self-supervised pretraining process. Specifically, we first update the model with the masked sensor reconstruction loss $\mathcal{L}_m$ for some epochs, because the convergence speed of $\mathcal{L}_m$ is slower than $\mathcal{L}_r$. Then, we update $\mathcal{L}_m$ and $\mathcal{L}_r$ together, i.e., $\mathcal{L}$, for more epochs until the loss does not decrease.

3.3 Fine-tune activity recognition

After self-supervised pretraining, we obtain a pretrained model $\mathcal{M}(., \Theta)$ that can extract feature representations of IMU samples. We then fine-tune the pretrained model with a subset of data with activity labels (D_{source}^L) from the source dataset to align with activity categories. To capture the temporal information, we employ a lightweight Transformer($\cdot$) module on top of the pretrained model.

$$o_i = \text{Transformer}(\mathcal{M}(., \Theta)(x_i)) \quad (14)$$

where $\mathcal{M}(., \Theta)$ is the pretrained model from our hierarchical self-supervised pretraining, and $\mathcal{M}(., \Theta)(x_i)$ represents extracting feature representation from the input IMU sample x_i with the pretrained model. x_i is an IMU sample from the target dataset D_{target} . Then, to interact features in the sample representation o_i and capture channel correlations, we employ a FeatureMixer($\cdot$) module.

$$\hat{y}_i = \text{FeatureMixer}(o_i) \quad (15)$$

where $\hat{y}_i$ is the predicted activity label. We implement the FeatureMixer($\cdot$) layer with a simple MLP layer. We use cross-entropy loss [90] as the loss function for the activity recognition. y_i is the ground truth activity label, e.g., *walking*.

$$\mathcal{L}_{Classification} = - \sum_{i=1}^{N_f} y_i \log(\hat{y}_i) \quad (16)$$

where N_f is the batch size in the fine-tuning process. Based on the loss $\mathcal{L}_{Classification}$, we apply Adam [37] to optimize the fine-tuning process until convergence.

$$\Phi_{updated} = \Phi - \eta \nabla_{\Phi} \mathcal{L}_{Classification} \quad (17)$$

Φ is the collection of all trainable parameters in the fine-tuning phase. η is the learning rate, $\nabla_{\Phi} \mathcal{L}_{Classification}$ is the classification loss in a batch, and $\Phi_{updated}$ is the updated parameters.

Through the integration of masked sensor reconstruction, contrastive regularization, and activity recognition fine-tuning, we obtain the trained model $\mathcal{M}(., \Phi)$, which is then evaluated with IMU data from target datasets D_{target} in the following experimental section.

4 EVALUATION

We evaluate the performance of CrossSHAR via extensive experiments on multiple real-world activity recognition datasets. We conducted extensive analysis to explore the following research questions.

- **RQ1 (Reliability):** What is the performance of CrossSHAR compared to state-of-the-art methods?
- **RQ2 (Scalability):** What is the performance of CrossSHAR when the scale of unlabeled training data increase?
- **RQ3 (Interpretability):** Whether important components of CrossSHAR contribute to the final performance?
- **RQ4 (Sensitivity):** What is the performance of CrossSHAR with different parameter values?
- **RQ5 (Feasibility):** What is the effectiveness and efficiency of CrossSHAR when deployed on mobile devices?

4.1 Evaluation setup

We introduce the evaluation setup, including evaluation metrics, dataset, preprocessing, baseline models, and implementation details.

4.1.1 Evaluation metrics. We employ average accuracy and F1-score to evaluate all models for multi-class activity classification. Specifically, all models are trained only with data from the source dataset and are evaluated with the data from the target dataset. Evaluation results on source datasets are explicitly mentioned in the experiments.

4.1.2 Dataset. We utilize UCI [56], Shoaib [63], Motion [46], and HHAR [65], to evaluate different methods, as they are widely used in previous HAR studies [7, 50, 58, 82, 83, 85]. Table 2 introduces the detailed statistics. UCI collects 6 activities (*downstairs, upstairs, walking, standing, sitting, and lying*) from 30 volunteers with a smartphone on the waist. Shoaib collects 7 activities (*downstairs, upstairs, walking, standing, sitting, jogging, and biking*) from 10 volunteers with smartphones. Motion collects 6 activities (*downstairs, upstairs, walking, standing, sitting, and jogging*) from 24 volunteers with smartphones in the front pocket. HHAR collects 6 activities (*downstairs, upstairs, walking, standing, sitting, and biking*) from 9 volunteers with smartphones and smartwatches.

Table 2. Dataset statistics.

Dataset	User	# of Activity	# of Sample	Device Type	Placement	Sampling Rate
UCI [56]	30	6	1,687	Smartphone	waist	50Hz
HHAR [65]	9	6	7,968	Smartphone, Smartwatch	-	50~200Hz
Shoaib [63]	10	7	7,500	Smartphone	left pocket, right pocket, wrist, upper arm, belt	50Hz
Motion [46]	24	6	4,108	Smartphone	front pocket	50Hz

4.1.3 Data preprocessing. To obtain consistent label space and feature space, we first select the important and commonly shared activities across these four datasets, i.e., *walking, standing or sitting, walking upstairs, walking downstairs*. Moreover, we further preprocess these datasets to get consistent sampling rates and window sizes. The sampling rate is set as 20 to simulate the resource-constraint scenario following the existing study [83]. We partition datasets according to the function of the dataset as source or as target. *For source datasets*, we partition each dataset as a training set (80%), validation set (10%) and test set (10%). To simulate the label scarcity setting, models utilizing unlabeled data are trained with the unlabeled training set and fine-tuned with only 10% of labeled training set data from the source dataset. Models utilizing only labeled data are trained with 80% of the labeled training set for a fair comparison. *For target datasets*, we take the whole dataset as the test set and directly use it to evaluate the cross-dataset performance.

4.1.4 Baselines. Our proposed model is compared against the 8 baseline models. These models are representative because they consist of pretraining-free models (MLP, CNN, ResNet, LSTM, and TSFCN), pretraining models (LIMUBERT and ContraTSC), and a state-of-the-art IMU domain generalization model (SDMix). We have not compared CrossHAR with another cross-dataset study [51] since the target dataset is unavailable during the training phase in our setting.

- **MLP [27]** (Multilayer Perceptron) is a classic deep learning model. It has received growing attention from time series [10] and computer vision [71] communities due to its simplicity and effectiveness in various tasks.
- **CNN [17]** (Convolutional Neural Network) is widely applied in many tasks across time series analysis and computer vision. CNN consists of four convolution layers and a pooling layer.
- **ResNet [24]** (Residual Neural Network) is a popular deep learning model and has been applied in time series analysis. ResNet addresses the degradation problem by utilizing residual connections to skip over certain layers. It has achieved state-of-the-art results in various tasks.

- **LSTM** [64] (Long Short-Term Memory) is a type of recurrent neural network architecture that is commonly used for time series classification tasks. LSTM captures the temporal correlation among different sampling points to generate a prediction of the activity category.
- **TSFCN** [78] (Time Series Fully Convolutional Network) has achieved very good performance in time series classification. The global average pooling layer is designed to identify the contribution region from IMU data.
- **LIMUBERT** [83] designs a self-supervised learning model to improve performance in activity recognition. The main component is the BERT-style pretraining task to learn useful IMU representation. The model has achieved very good performance in classifying human activities.
- **ContraTSC** [16] employs contrastive learning as the pretraining task to learn robust representation for time series classification. It has achieved good performance in multiple downstream time series classification tasks.
- **SDMix** [42] is a state-of-the-art cross-domain activity recognition model. The core design is a Semantic-Discriminative Mixup approach that overcomes the semantic inconsistency brought by domain gaps across a single domain or dataset. SDMix improves Mixup [87], a popular data augmentation method that enlarges training data diversity, by introducing semantics and a large margin loss in the design.

4.1.5 Implementation details. Our framework was developed on a system equipped with 40GB of RAM and a P40 GPU with 24GB of memory. All models were implemented with Python 3.8 and PyTorch 1.12. In the pretraining phase, we set the batch size to 512 and the number of training epochs to 150, with a learning rate of 0.001. We applied a masking ratio of 15% to each channel across all IMU samples. For the fine-tuning stage of the model, the batch size was set to 128, and the learning rate was set to 0.001. The model's architecture for fine-tuning included a single-layer transformer with 4 attention heads. To ensure robustness and optimal performance, all deep learning-based baseline models are trained for a minimum of 150 epochs. This training duration was determined as no significant improvements in performance were observed beyond this point.

4.2 Overall results (RQ1)

In Table 3, we present the overall cross-dataset performance of CrossHAR compared with multiple models on four datasets. Note that for each model, we train them on four source datasets separately and evaluate them on corresponding target datasets. We then report the detailed results in each activity category by confusion matrices. In Table 4, we present the performance of the non-transfer setting. Due to the space limit, we only present the average accuracy while CrossHAR also achieves the highest average F1 score compared to SoA models.

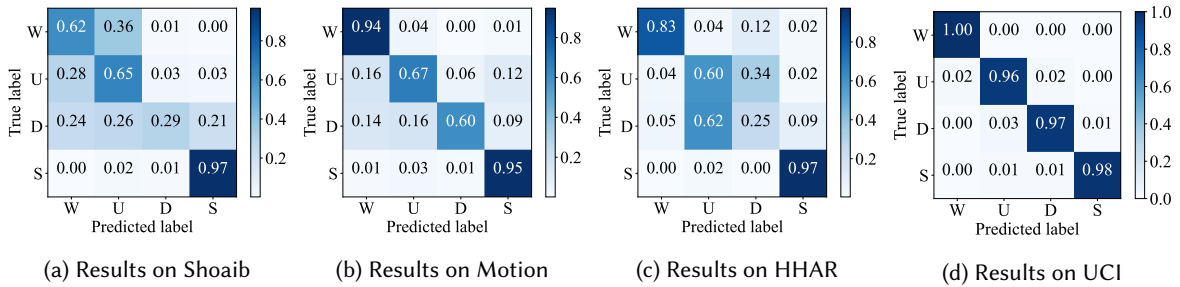
4.2.1 Results of one-to-many cross-dataset setting. By analyzing the results in Table 3, we have multiple main observations. **(i) CrossHAR achieves the best performance.** Overall, our CrossHAR achieves the best performance across all target datasets trained with different source datasets. More importantly, CrossHAR outperforms the second-best model by 10.83% in average accuracy. These results demonstrate the strong generalization capability of CrossHAR. **(ii) Domain generalization and pretraining.** On average, the second-best model is shared by SDMix and LIMUBERT (62.70%), which is a state-of-the-art domain generalization model and a novel IMU pretraining model, respectively. The success of SDMix demonstrates the advantage of the mixup-based sensor data augmentation. As a state-of-the-art pretraining method, LIMUBERT outperforms ContraTSC significantly. This result demonstrates that modeling the local pattern of the IMU sample brings better generalization capability. **(iii) Non-symmetric transferability across different datasets.** We found that in cross-dataset recognition, a model trained on Dataset A excels when applied to Dataset B, but a model trained on Dataset B does not maintain this performance on Dataset A; instead, it performs best on Dataset C. For instance, for the HHAR dataset, the best performance is achieved when transferring to the UCI dataset. For UCI, however, the best performance is achieved when transferring to the Motion dataset. This indicates that the transferability of learned representations may be

Table 3. Overall evaluation result (Accuracy) of the cross-dataset setting. The **bold** and underline represent the best and second-best results.

Source	Target	MLP	CNN	ResNet	LSTM	TSFCN	LIMUBERT	ContraTSC	SDMix	Ours
UCI	Shoaib	46.74	49.38	50.51	32.5	36.45	<u>58.28</u>	51.01	57.67	69.93
	Motion	53.08	43.68	52.49	64.75	46.19	<u>68.94</u>	52.29	66.18	87.90
	HHAR	45.94	56.46	51.21	68.15	44.81	<u>69.44</u>	42.86	61.10	77.27
Shoaib	UCI	60.64	64.90	66.53	59.5	70.37	53.00	55.36	71.77	<u>67.81</u>
	Motion	58.59	72.05	59.18	73.78	60.03	69.50	53.55	65.07	<u>72.18</u>
	HHAR	50.18	59.50	58.56	57.45	<u>60.57</u>	53.00	42.86	58.31	67.03
Motion	UCI	56.43	50.06	63.52	53.31	57.51	<u>72.26</u>	45.05	65.09	76.41
	Shoaib	57.21	56.92	60.37	56.82	55.74	<u>65.72</u>	43.21	59.93	70.89
	HHAR	52.37	43.12	53.49	43.7	49.62	<u>64.63</u>	42.86	74.09	<u>65.12</u>
HHAR	UCI	54.51	48.85	48.02	40.5	48.02	<u>67.04</u>	55.54	54.49	91.52
	Shoaib	53.89	24.39	44.88	19.45	41.14	53.56	26.24	<u>57.49</u>	61.08
	Motion	60.28	60.86	51.64	37.11	64.38	57.04	<u>65.87</u>	61.14	75.17

influenced by the inherent characteristics and distributions of the datasets involved, leading to varying levels of performance depending on the source and target datasets.

4.2.2 Results of each category. Fig. 7 presents the detailed activity recognition results with confusion matrices. Fig. 7d exhibits the best performance, which is because we present the result trained on the UCI dataset. Note that in the cross-dataset recognition, *downstairs* is the most difficult activity to be recognized according to Fig. 7a, Fig. 7b, and Fig. 7c, which is because the pattern of IMU data in *downstairs* is very similar to *upstairs*.

Fig. 7. Confusion matrix on four target datasets. W, U, D, S represent *walking*, *upstairs*, *downstairs*, and *still*, respectively.

4.2.3 Results of non-transfer setting. To evaluate the performance of different methods in the source dataset, i.e., the training and testing data are both from the same dataset. The training, evaluation, and testing partition follows the same setting in the source dataset. Results are shown in Table 4. Note that most models except MLP achieve comparable performance. On average, TSFCN achieves the best performance in this non-transfer setting. Results from Table 3 and Table 4 show that CrossHAR outperforms others significantly in the cross-dataset transfer scenario while maintaining similar performance in the non-transfer setting.

Table 4. Overall evaluation result of non-transfer setting.

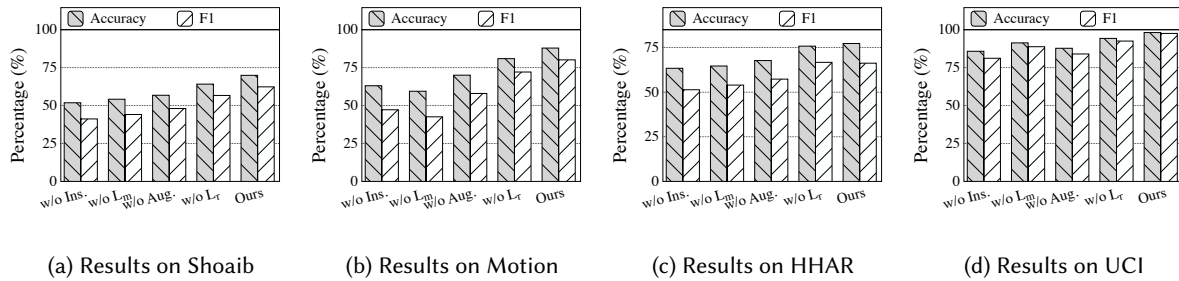
Model	MLP	CNN	ResNet	LSTM	TSFCN	LIMUBERT	ContraTSC	SDMix	Ours
UCI $\rightarrow$ UCI	74.74	100	100	88.8	100	87.56	87.06	99.11	98.12
Shoaib $\rightarrow$ Shoaib	80.33	98.25	98.90	92.28	100	90.78	94.53	98.87	95.07
Motion $\rightarrow$ Motion	78.99	98.61	99.13	94.53	99.48	90.67	91.50	98.78	93.59
HHAR $\rightarrow$ HHAR	90.15	99.05	99.91	96.09	100	92.17	98.87	97.87	99.20

Table 5. Overall evaluation result of multi-dataset training.

Model	MLP	CNN	ResNet	LSTM	TSFCN	LIMUBERT	ContraTSC	SDMix	Ours
HMS $\rightarrow$ UCI	67.43	67.19	65.87	70.19	66.77	61.59	54.3	75.72	88.68
HMU $\rightarrow$ Shoaib	64.54	63.09	66.29	38.98	67.71	58.0	49.0	70.09	73.67
HSU $\rightarrow$ Motion	70.12	77.15	59.28	54.57	61.21	63.70	52.24	66.17	78.26
MSU $\rightarrow$ HHAR	61.47	58.71	56.75	42.97	51.08	63.26	43.36	74.44	76.19

4.3 Pretraining on multiple source datasets (RQ2)

To further evaluate the utility brought by large-scale unlabeled data, we design the many-to-one training paradigm. We assume that during the model training phase, we have access to multiple unlabeled datasets as the source datasets, which might generate a model with better generalization capability on the target dataset. We represent UCI as U, Shoaib as S, Motion as M, and HHAR as H. From Table 5, we can observe that our method achieves the highest accuracy in all datasets. Overall, the average accuracy of our model increases by 5.67%. Also, with more training data, most models trained with multiple source datasets outperform models trained with a single source dataset. For example, when trained with one source dataset, SDMix achieved 64.13% average accuracy on the Motion dataset. The accuracy is increased to 66.17% with the help of more source training datasets. For straightforward baseline models such as MLP, the performance is also improved with the scale of training data. Therefore, these results demonstrate that one potential approach to addressing the domain shift challenge is to increase the training datasets.

Fig. 8. Impact of **different components** on the cross-dataset performance of CrossHAR.

4.4 Impact of different components (RQ3)

To further study the importance of different components, we conduct extensive evaluations and report the corresponding results. To save space, if not otherwise specified, the results are conducted by pretraining and fine-tuning with the UCI dataset, and then evaluated on itself and other datasets.

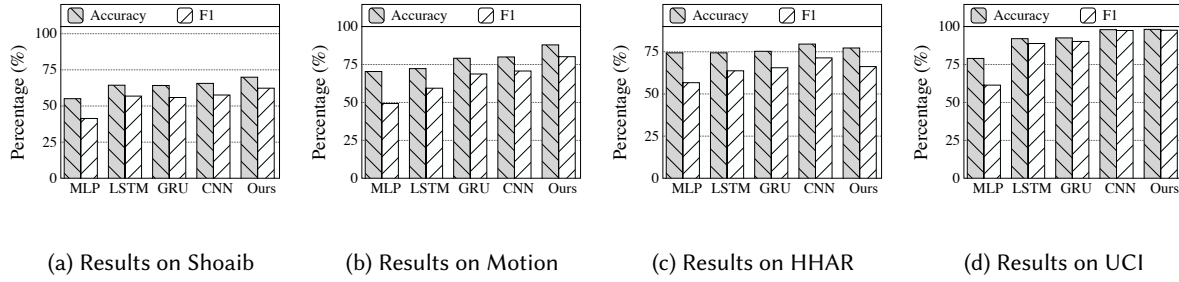


Fig. 9. Impact of **different model structures** in fine-tuning step on the cross-dataset performance of CrossHAR.

4.4.1 Impact of normalization. We evaluate the impact of sensor normalization during pretraining (w/o Ins). As shown in Fig. 8, the accuracy drops by 18.09%, 24.83%, and 13.89% on Shoaib, Motion, and HHAR datasets, respectively. Thus, the results demonstrate that normalizing IMU data with instance normalization improves the generalization capability significantly.

4.4.2 Impact of masked sensor pretraining. We evaluate the impact of the masked sensor pretraining by pretraining and fine-tuning CrossHAR without $\mathcal{L}_m$ and keeping others the same (w/o $\mathcal{L}_m$). As shown in Fig. 8, the accuracy drops by 15.76%, 28.46%, and 12.62% on Shoaib, Motion, and HHAR datasets, respectively. Such a significant performance drop demonstrates the importance of $\mathcal{L}_m$ in capturing domain-invariant representations.

4.4.3 Impact of physically-informed data augmentation. We evaluate the impact of the physically-informed sensor data augmentation by removing this module and keeping others the same. As shown in Fig. 8, removing the physically-informed data augmentation module (w/o Aug) brings a significant performance drop across all datasets. The accuracy drops by 13.14%, 17.87%, and 9.56% on Shoaib, Motion, and HHAR datasets, respectively. Similarly, the F1 drops by 14.29%, 22.13, and 8.93% on the Shoaib, Motion, and HHAR datasets, respectively.

4.4.4 Impact of contrastive regularization. We evaluate the impact of the contrastive regularization by pretraining and fine-tuning CrossHAR without $\mathcal{L}_r$ and keeping others the same (w/o $\mathcal{L}_r$). As shown in Fig. 8, the accuracy drops by 5.8%, 6.98%, and 1.48% on Shoaib, Motion, and HHAR datasets, respectively. These results demonstrate the importance of the regularization process during the self-supervised pretraining.

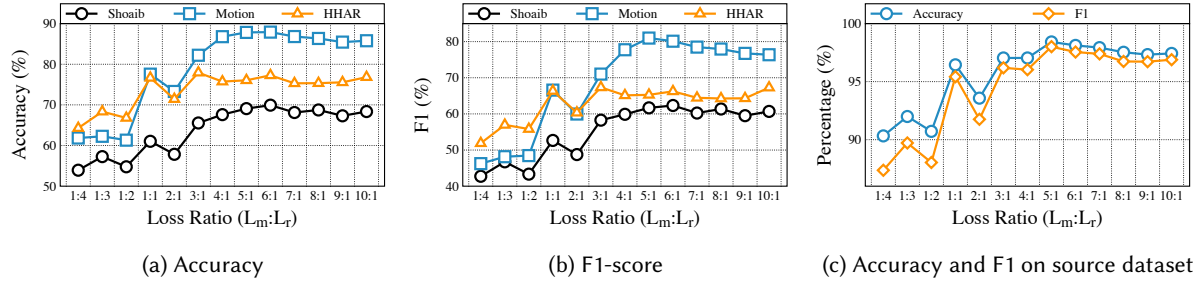
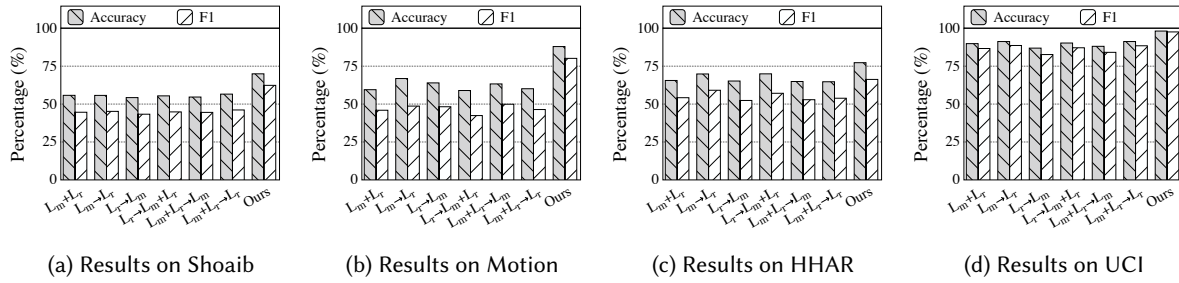
4.4.5 Impact of fine-tuning model structure. We evaluate the impact of different model structures in the fine-tuning step by replacing our Transformer with MLP, LSTM, GRU, and CNN. As shown in Fig. 9, Transformer achieves the best performance in most cross-dataset cases and MLP has the lowest performance. For the non-transfer setting in Fig. 9d, CNN achieves similar performance with Transformer, higher than other methods.

4.5 Sensitivity analysis (RQ4)

We evaluate the sensitivity of important parameters. The most important parameters include parameters α and β determining the loss ratios. We also evaluate the performance variance with different pretraining paradigms. If not otherwise specified, the results are conducted by pretraining and fine-tuning with the UCI dataset, and then evaluated on itself and other datasets.

4.5.1 Impact of the ratio between two losses. In the self-supervised pretraining process, an important factor that impacts the model performance is the ratio between the masking loss and regularization loss, i.e., values of α and β . To evaluate its impact, we pretrain our model with varying $\alpha : \beta$ from 1:4 to 10:1. Results are shown in Fig. 10.

From Fig. 10, we can observe that when the weight of the $\mathcal{L}_m$ (α) is very small, the model's ability to recognize human activities is significantly impaired. As α increases, the performance of the model progressively improves.

Fig. 10. Impact of the **loss ratio** $\alpha : \beta$ ($\mathcal{L}_m : \mathcal{L}_r$) on cross-dataset performance.Fig. 11. Impact of **self-supervised training paradigm** on the cross-dataset performance of CrossSHAR.

The average performance yields the highest across all datasets when $\alpha : \beta$ is 6:1, further increasing α decreases the activity recognition performance. Note that CrossSHAR has a similar F1 score at 5:1 and 6:1, we select 6:1 as the optimal ratio because it brings the highest accuracy for more datasets.

4.5.2 Impact of pretraining paradigm. We evaluate the impact of different training paradigms during self-supervised pretraining and show the results in Fig. 11. Specifically, given two loss functions in the pretraining process, there are 7 ways to update our model according to the updating order. (i) update $\mathcal{L}_m$ and $\mathcal{L}_r$ simultaneously ($\mathcal{L}_m + \mathcal{L}_r$), (ii) update $\mathcal{L}_m$ and then update $\mathcal{L}_r$ ($\mathcal{L}_m \rightarrow \mathcal{L}_r$), (iii) update $\mathcal{L}_r$ and then update $\mathcal{L}_m$ ($\mathcal{L}_r \rightarrow \mathcal{L}_m$), (iv) update $\mathcal{L}_r$ and then update $\mathcal{L}_m$ with $\mathcal{L}_r$ together ($\mathcal{L}_r \rightarrow \mathcal{L}_m + \mathcal{L}_r$), (v) update $\mathcal{L}_m$ with $\mathcal{L}_r$ together and then update $\mathcal{L}_m$ ($\mathcal{L}_m + \mathcal{L}_r \rightarrow \mathcal{L}_m$), (vi) update $\mathcal{L}_m$ with $\mathcal{L}_r$ together and then update $\mathcal{L}_r$ ($\mathcal{L}_m + \mathcal{L}_r \rightarrow \mathcal{L}_r$), and (vii) update $\mathcal{L}_m$ and then update $\mathcal{L}_m$ with $\mathcal{L}_r$ together ($\mathcal{L}_m \rightarrow \mathcal{L}_m + \mathcal{L}_r$, ours).

As shown in Fig. 11, our training paradigm achieves the best performance across all datasets. The second-best training paradigm is updating $\mathcal{L}_m$ and then update $\mathcal{L}_r$ ($\mathcal{L}_m \rightarrow \mathcal{L}_r$). These results follow our assumption that firstly pretrain the model with the loss that is hard to converge can improve the model performance. Thus, training with the masked sensor modeling loss $\mathcal{L}_m$ first and then incorporating the contrastive regularization loss $\mathcal{L}_r$ achieves the best performance. Our finding could benefit broader directions involving self-supervised pretraining with multiple objective functions in the ubiquitous computing community.

4.6 Deployment on mobile devices (RQ5)

In this subsection, we focus on the practical case study of our HAR model, CrossSHAR, by deploying and testing it on a smartphone. In this case study, we aim to evaluate the performance of CrossSHAR against new sensors (some public datasets were collected a decade ago [56]), under diverse conditions such as various motion amplitudes, and in limited hardware resources. We present the process of the case study in Fig. 12(a).

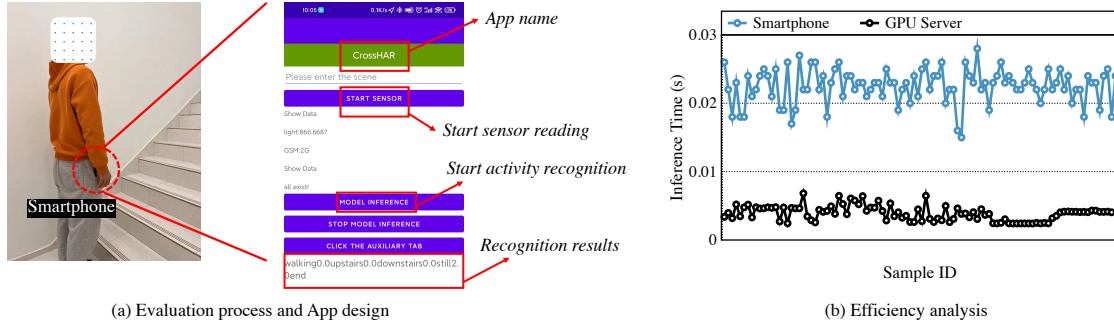


Fig. 12. Illustration of the (a) case study process and our activity recognition application design, and (b) efficiency analysis of CrossHAR when deployed on a smartphone.

Setup. The deployment of CrossHAR on mobile devices consists of three main steps. Initially, we developed an Android application using Android Studio [2]. Subsequently, CrossHAR was packaged into a *pt* file, which was then loaded into the application. The app was designed to read data from the accelerometer and gyroscope sensors of the device and feed this data to the loaded model for activity recognition. For our case study, we selected the Redmi K30 Pro smartphone, equipped with an Android 11 operating system, 8GB of memory, and a Qualcomm Snapdragon 865 CPU. The *pt* file model size loaded in the app is 3.84MB.

Evaluation. To evaluate the model's performance in a real-world scenario, we conducted a preliminary case study with five volunteers ranging from 20 to 30 years old. Each volunteer performed four activities, *still*, *walking*, *upstairs*, and *downstairs*, with 2 minutes of data collected for each activity. The model for deployment is trained with the UCI dataset [56]. The model provided activity recognition predictions at a rate of one prediction per second. The activity recognition accuracy for five volunteers was 69.79%, 66.88%, 55.65%, 60.7%, and 60.64%, respectively. There are two main reasons for the relatively low accuracy compared to the results on public datasets. Firstly, in our setting, volunteers conducted activities with strong intensity or low intensity. For example, slowly walking upstairs is easy to recognize as walking or still. Therefore, their performances are lower (55.65%, 60.7%, and 60.64%). Secondly, the class for each activity is more balanced compared to public datasets. These results highlight substantial opportunities for enhancement in future studies.

In terms of inference speed, the average time taken per sample was 0.0225 seconds, with a minimum and maximum of 0.015 seconds and 0.028 seconds, respectively. For comparison, we also assessed the model's inference efficiency on a single P40 GPU with a batch size of 1. The average inference time for one IMU sample on the GPU was 0.00389 seconds, with a minimum of 0.00241 seconds and a maximum of 0.00689 seconds. The detailed efficiency analysis is presented in Fig. 12(b). While the inference speed on the smartphone was approximately 5.78 times slower than on the GPU server, it remains within a practical range for real-world applications. This demonstrates that CrossHAR can effectively function on mobile devices, albeit with a reduced inference speed compared to GPU-based servers.

5 DISCUSSION

In this section, we summarize some lessons learned according to our in-depth data-driven analysis and extensive experimental results. We also discuss the limitations and a few future directions.

5.1 Lessons learned

- **Heterogenous domain shifts across datasets.** We conduct systematic data-driven analysis and reveal that the domain shifts across different datasets are significant (see Fig. 2 and Fig. 4). Moreover, it is more severe

than the domain shift across a single domain (see Fig. 1). To increase the effectiveness of HAR models when deployed in the real world, more efforts should be spent to address such heterogeneous domain shifts.

- **Physically-informed sensor data augmentation.** We show that motivated by the principles of the IMU data generation process, we can design a strong augmentation method that increases the diversity of source data and improves the generalization capability across heterogeneous domain shifts (see Fig. 8).
- **Self-supervised pretraining for mobile sensing.** Self-supervised pretraining can increase the generalization ability of HAR models significantly. We demonstrate that via a proper design of the model structure and pretraining paradigm, the model can learn a domain-invariant representation of mobile sensing data purely from unlabeled data from source domains (see Fig. 3).

5.2 Limitation and future work

While our physically-informed data augmentation and hierarchical self-supervised pretraining methods offer a promising approach to generalize human activity recognition, we identify two limitations. Firstly, we only utilize IMU data from a few public datasets for pretraining. It is still unclear whether the generalization ability can be further improved by increasing the scale of unlabeled IMU data. Also, while there are many augmentation methods in time series, computer vision, and natural language processing, it is unclear whether the incorporation of more augmentation approaches can improve recognition performance. Therefore, we would like to explore the following directions in future studies.

- **Towards foundation models for human activity recognition.** Motivated by the success of the foundation model in NLP [69] and computer vision communities [38], we will explore developing foundation models for sensor-based human activity recognition [40]. Building upon CrossHAR, we would like to improve the sensor data pretraining from the perspective of increasing the pretraining scale and the perspective of designing advanced pretraining paradigms for sensor data.
- **Diverse data augmentation design.** Many time series augmentation methods have been explored including basic approaches and advanced approaches like utilizing deep generative models [79]. Moreover, Generative Adversarial Networks (GANs) and diffusion models are also receiving growing attention for sensor data generation [5, 91]. Proper integration of these augmentation methods could potentially improve the generalization capability of activity recognition models.
- **Evaluation and performance improvement.** Due to the limited number of shared activities in our datasets, we only evaluate the performance of a few activities. We will increase the number of activities in future studies. Although CrossHAR demonstrates significant superiority over state-of-the-art methods, there is still ample room for further enhancement. We will continue to improve cross-dataset recognition in future studies.

6 RELATED WORK

6.1 Wearable-based human activity recognition

With the widely deployed wearable devices such as smartphones and smartwatches, the ubiquitous sensors on these devices provide a unique opportunity for HAR [3, 7, 9, 18, 21, 45, 48, 61, 66, 86]. Chen et al. [8] provide a systematic review of sensor-based HAR. Except for regular activities, activities in specialized settings are also studied [26, 54, 81]. To increase the safety and robustness of HAR models, Jeyakumar et al. [30] design an explainable complex human activity recognition model for critical applications such as healthcare. Li et al. [41] study the fairness aspect of HAR models. Recently, with the increase of large-scale unlabeled data, weak-supervised, self-supervised, and unsupervised methods are receiving more attention [22, 23, 29, 58, 62, 68, 83, 86]. Given the intrinsic similarity of wearable sensor data and other data such as video, some studies focus on HAR by utilizing knowledge learned from other modalities [4, 39, 55, 72]. Bhalla et al. [4] explore utilizing IMU data

with mmWave radar for HAR and have achieved good performance. Kwon et al. [39] and Santhalingam et al. [60] extract IMU data from recorded videos to improve HAR performances.

6.2 Generalize human activity recognition

Due to the heterogeneity of sensor readings, models trained on one domain might not achieve comparable performance on other domains [49, 51, 83]. Therefore, some recent studies focus on the generalization capability of HAR models. Presotto et al. [49] combine multiple public datasets to improve HAR performance. Qin et al. [51] design an adaptive spatial-temporal transfer learning model to select the most proper source domain data and improve the transfer performance on different datasets. Hu et al. [25] harness unsupervised domain adaption for cross-person HAR. Taking one step further, some methods have been recently designed to train HAR models only with data from source domains [42, 43, 52, 84, 88, 89], i.e., domain generalization [35, 76]. Lu et al. [43] propose pseudo domain-class labels and adversarial self-supervised pseudo labeling to learn domain-invariant representation. Qin et al. [52] design a fusion method to fuse domain-invariant and domain-specific representations thereby increasing the generalization capability. To address the shortage of training data, Qin et al. [53] design a diverse and discriminative representation learning model to generalize HAR to unseen domains. Motivated by the practical value of model generalization, Xu et al. [84] conduct a systematic evaluation of cross-dataset depression detection and propose two new algorithms to improve the generalization performance.

6.3 Self-supervised pretraining

Recently, self-supervised pretraining has received increasing attention from the research community due to its great success in natural language processing [32] and computer vision [14, 19, 33]. It has been demonstrated empirically to benefit downstream applications by learning generalizable knowledge from unlabeled data [31]. Recently, some self-supervised pertaining methods over time series data have been proposed [15, 16, 20, 44]. Two representative self-supervised learning algorithms are masked language modeling [32] and contrastive learning [11, 88]. Xu et al. [83] design a pretraining paradigm based on masked language modeling, and have achieved good performance in HAR. For time series classification, Eldele et al. [16] design a contrastive learning algorithm to learn robust embeddings of time series data.

Summary. In summary, this study is different from existing work because of the following aspects.

- Firstly, we aim to generalize HAR from the source dataset to the target dataset that shares a combination of multiple domain differences, while most existing studies focus on settings without domain shift or with domain shift only in one domain.
- Secondly, we focus on HAR without any labeled or unlabeled training data from the target dataset, which is more challenging and different from most existing studies.
- Moreover, to our knowledge, most of the existing self-supervised methods are not designed for IMU sensor data in cross-dataset HAR. We innovatively design a hierarchical self-supervised pretraining method to learn generalizable representations from unlabeled IMU data.

7 CONCLUSION

In this paper, we study the problem of cross-dataset human activity recognition without any data from the target dataset in the model training phase. Specifically, we design CrossHAR, a novel model for generalizable activity recognition. CrossHAR consists of three main components. (i) A Physically-informed Sensor Data Augmentation approach to increase the data diversity in the source dataset. (ii) A Hierarchical Sensor Data Pretraining approach is designed to learn generalizable representation based on the augmented sensor data. (iii) A Fine-tune Activity Recognition approach to align the pretrained model with activity categories. We conduct extensive experiments

on multiple real-world datasets with systematic analysis. Results show that CrossHAR outperforms SoA models significantly in cross-dataset activity recognition.

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